

MID TERM EXAMINATION

Course: Data Structures (CS101)

Set: C | Time: 3 Hours | Max Marks: 128 | Date: 15-03-2026

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

GENERAL INSTRUCTIONS:

1. Answer all questions.
2. Figures to the right indicate full marks.

SECTION A: MCQs

1. Which of the following is NOT explicitly mentioned as a type of scenario used in architectural evaluation? [3 Marks]

- a) Use-case scenarios
- b) Performance scenarios
- c) Growth scenarios
- d) Exploratory scenarios

2. A direct consequence of using an off-the-shelf component is that the architecture must primarily: [3 Marks]

- a) Change the component's internal behavior.
- b) Adjust to the component's way of working.
- c) Eliminate vendor dependency.
- d) Increase customization options.

3. In design decision-making, which factor is guided by what the system must do, such as supporting payments or product search in an e-commerce system? [3 Marks]

- a) Non-Functional Requirements
- b) Stakeholder Needs
- c) Functional Requirements
- d) Technology Constraints

SECTION B: SHORT ANSWERS

4. Name and briefly explain two types of scenarios mentioned for architectural evaluation. [6 Marks]

5. Explain how using off-the-shelf components can lead to "extra integration work" and "limited customization"**[6 Marks]**
6. According to the text, what role do "Technology Constraints" and "Team Skills and Experience" play in design decision-making?**[6 Marks]**
7. List four key stakeholders whose participation is required for successful architectural evaluation.**[6 Marks]**
8. Describe two negative impacts of off-the-shelf components related to vendor interaction and future changes.**[6 Marks]**
9. Briefly explain the significance of "Risk and Uncertainty" and "Reusability and Extensibility" as factors for design decision-making.**[6 Marks]**
10. What are "Risks," "Sensitivity points," and "Trade-offs" in the context of successful architectural evaluation?**[6 Marks]**

SECTION C: LONG ANSWERS

11. Elaborate on the importance of "Clearly Defined Quality Attributes" and "Well-Specified Scenarios" for a successful architectural evaluation, providing examples as given in the text.**[11 Marks]**
12. Discuss the challenges and limitations introduced by using off-the-shelf components, focusing on how the architecture adapts and potential quality attribute impacts.**[11 Marks]**
13. Explain how "Functional Requirements," "Non-Functional Requirements," and "Stakeholder Needs" collectively guide design decision-making in software architecture.**[11 Marks]**
14. Describe the role of "Structured Evaluation Methods" and an "Experienced Evaluation Team" in achieving successful architectural evaluation, including examples of methods and team expertise.**[11 Marks]**
15. Analyze how "Budget and Schedule," "Team Skills and Experience," and "Technology Constraints" interact as practical factors influencing design decisions.**[11 Marks]**
16. Drawing from the text, explain how the "Impacts of Off The Self Components on Architecture" might influence "Factors for Design Decision-Making in Software Architecture," specifically considering budget/schedule and customization aspects.**[11 Marks]**
17. Discuss the comprehensive nature of successful architectural evaluation by explaining how "Architectural Documentation," "Proper Tools and Techniques," and "Good Communication" contribute to the process.**[11 Marks]**
-

--- End of Question Paper ---